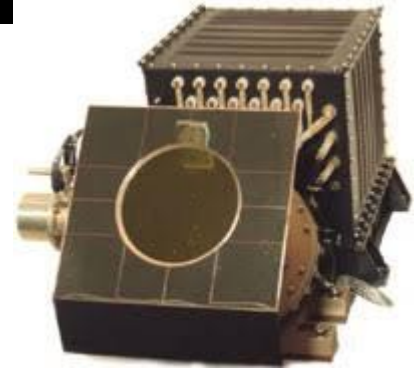


Machine Learning Methods in Predicting High Energy Protons in L1 Region

Lewis Adkins

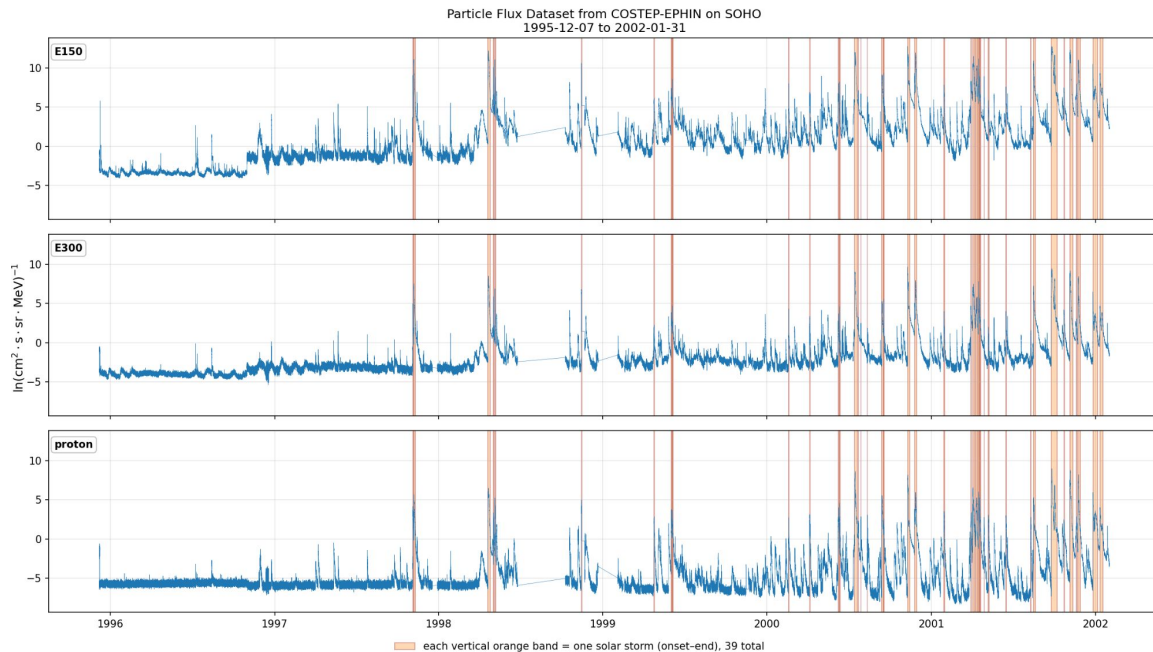
Intro

- Replicating [Torres 2025](#) paper that tests Neural Networks and Recurrent Neural Networks to predict high energy protons SEPs
- Data collected on the COSTEP-EPHIN instrument onboard SOHO
- Located in the L1 region



Data

- Two electrons channels ($>.25$ and $.67$)MeV and a proton channel >10 MeV
- December 1995 - January 2002
- 5 minute cadence
- 39 Identifiable proton events



Torres's Machine Learning Methods

- Torres sought to improve proton predictions attempted [Posner](#) who used a electron parameter forecasting matrix. With the advent of machine learning method, RNN and NN were proposed as more efficient method for forecasting
- **Goal** : By taking in a window of from (t-24, t) of e_{low} , e_{high} , and current proton flux can we reliably predict SEP storms 30 minutes (t+6) and 60 minutes (t+12) out.

Parameters

- For each intensity model a single hidden layer of 30 units is used using sigmoid activation.
- Weights updated form adam optimizer.
- Each are run 5 times with different random initializations .
- early stopping if no change larger than 10^{-4} after 20 iterations
- Data split 80-20 for training and testing respectively.
 - From the split, 21 SEP events fall into the training set and 18 SEP events fall into the testing set

Small models -
 d_{model} , dim_{val} ,
 $attn_heads$,
 $n_encoders$

Models	Parameters
NN	2,311
RNN	3,181
Sinusoidal	3,361
Zero Init	3,761
RoPE	3,361

Figure 4. Model type and their number of total parameters

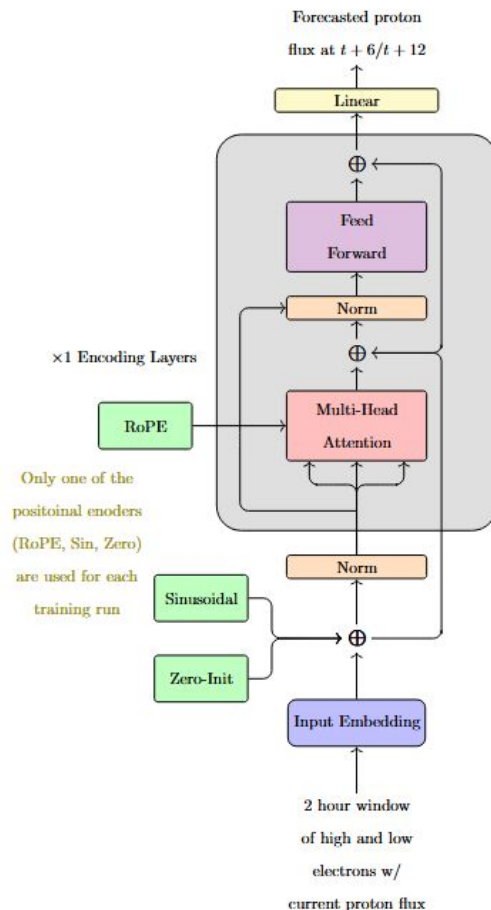
Can Transformers be Better?

From [Vaswani 2017](#) transformer have been used in many fields to make predictions

The types of transformer in this work has been identified by the positional encoding method (RoPE, Sin, Zero-Init) and other well known packages such as Autogluon

Transformers Tested

- Sinusoidal
- Rope
- Zero-Init
- Autogluon Temporal Fusion Transformer
- Autogluon Chronos 2



Sinusoidal

Sinusoidal positional encoding was first introduced in the original proposed trans-former architecture paper (Vaswani et al., 2017). It is an absolute position encoding method that uses sin and cos functions of different frequencies to encode the positional indexes of tokens in a sequence. For each token, a vector of dimension dim val is given. each token vector is determined by the position they are in, in a sequence and the length of the vector. Add and even indexes in the vector are from (2) and (1) respectively. Once these tokens are indexed, the attention block computes their own QKV matrix allowing for predictions

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{model}})$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{model}})$$

The sin and cos functions keep all values tokenized values within a range of $[-1, 1]$ which is easy for the transformer to understand eliminating the need for standardization. The positions are encoded uniquely based on value and position, allowing it to scale to unlimited sequence lengths.

RoPe

Rotary Positional Embedding (RoPE) is a relative positional embedding method that encodes the relative positions of embedded vectors to the dot product of two rotated vectors. The rotation of the vectors is determined by their order in the sequence. The relative positions are then used for self attention formulation (Su et al., 2021). The rotated embedded vector is the product of the rotation matrix, query-key weight matrix and our input vector

$$f_{\{q,k\}}(\mathbf{x}_m, m) = \begin{pmatrix} \cos m\theta & -\sin m\theta \\ \sin m\theta & \cos m\theta \end{pmatrix} \begin{pmatrix} W_{\{q,k\}}^{(11)} & W_{\{q,k\}}^{(12)} \\ W_{\{q,k\}}^{(21)} & W_{\{q,k\}}^{(22)} \end{pmatrix} \begin{pmatrix} x_m^{(1)} \\ x_m^{(2)} \end{pmatrix}$$

A property of RoPE is long term decay. This means that pairs of tokens at large relative distance will have less connection than those that are close to each other (Su et al., 2021). For our flux data, this would coincide with the idea from (Posner, 2007) that only a short term forecasting of relativistic electrons is required to accurately predict future proton storms from 30 minutes to an hour

Zero Init

Zero-Initialized positional encoding is an absolute positional that sets all positional encoding parameters in a transformer to zero at the start of training. This means that position of a timestamped flux has equal effects on each other at the start. Then as training continues, near timestamped fluxes learn to correlate to each other rather than further ones. With a zero or near zero initialization, models on their own learn patterns that are used to make predictions with zero valued starting weights.

This makes it reliable for generalization which is ideal for SEP data made almost entirely of background fluxes (Torres et al., 2025). Commonly, small near zero initializations sampled with normal distributions were found to perform better than larger ones (Ito et al., 2024).

Most timestamps in the relatively small proton electron flux dataset are near zero background values. A larger starting weight already has background information included leading to forecasting more background in the future during the early portion of training. It's not until after rounds of epochs to unlearn patterns are to finally map where spikes in fluxes start to occur in order to make accurate predictions.

Auto Gluon

Only 2 transformers out of the library of forecasting transformers were applicable to the problem set. ***Supports_past_covariates*** = True can only be set for Temporal Fusion Transformer (TFT) and the new Chronos2 model. The e_low, e_high are past covariates as they are known historically. If set to false, the model can no longer accept our (t-24, t) window of all three e_low, e_high and pronto channels and instead just uses the proton channel to make predictions

Model	Known covariates	Past covariates	Verdict
TemporalFusionTransformer	Y	Y	✓ qualifies
Chronos-2	Y	Y	✓ qualifies
DeepAR	Y	N	proton-only
PatchTST	Y	N	proton-only
TiDE	Y	N	proton-only
WaveNet	Y	N	proton-only

Training

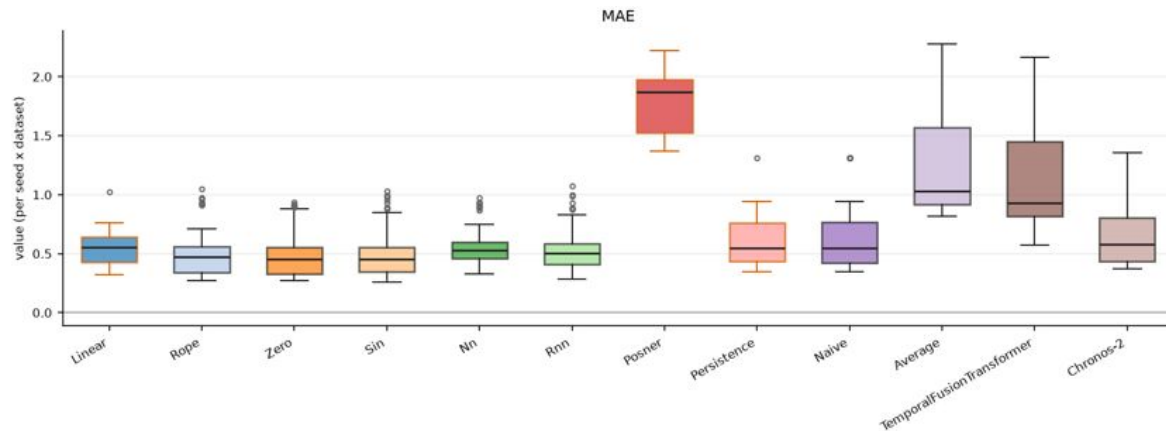
- Replicating to the fullest extent the size of the models for adequate comparison (early stopping, d_model, dim_val, adam_optimizer etc)
- Bootstrapped creating 9 other new datasets (10 in total) of resampled data from 1995-2002. Each dataset now gets 10 seeds instead of 5. 100 models for each model type
- Baseline models that are non machine learning:
 - Linear regression
 - Persistence
 - Autogluon average
 - Autogluon Naive
 - Posner

Metrics

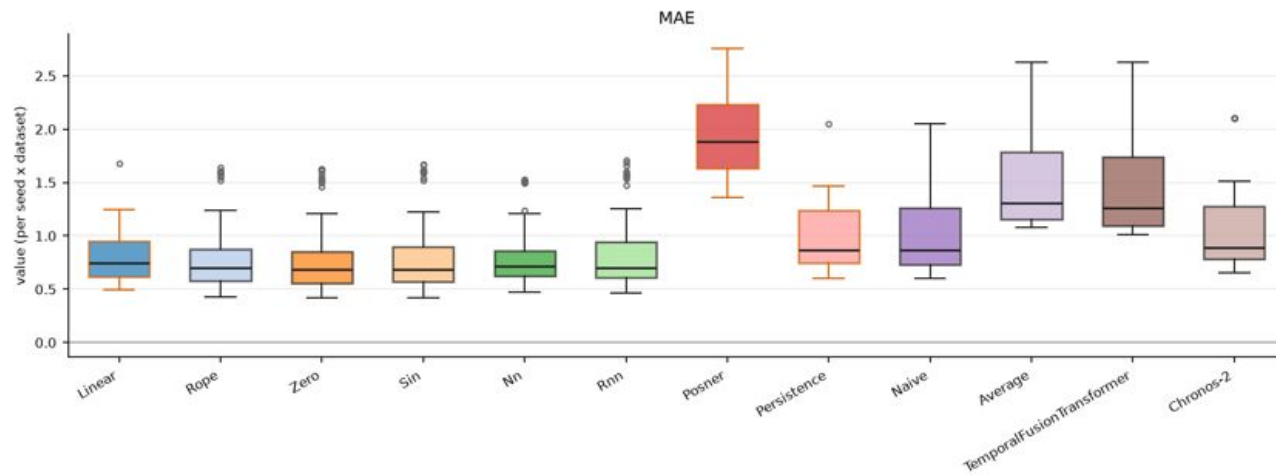
To judge predictions we use the following statistics:

- Time
- MAE
- PE
- Lag
 - Ln10
 - Onset to peak o2p lag
 - 02t onset to threshold
- F1

Results - MAE,

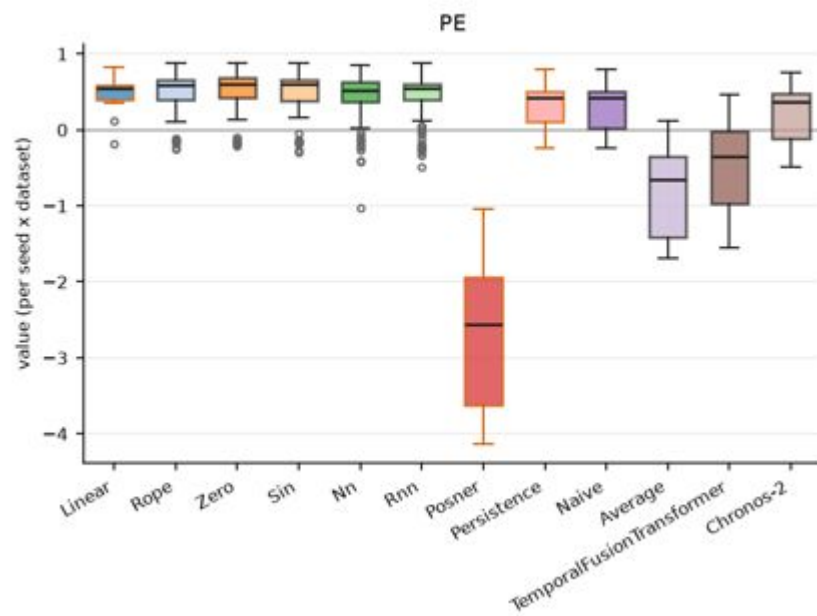


t+12

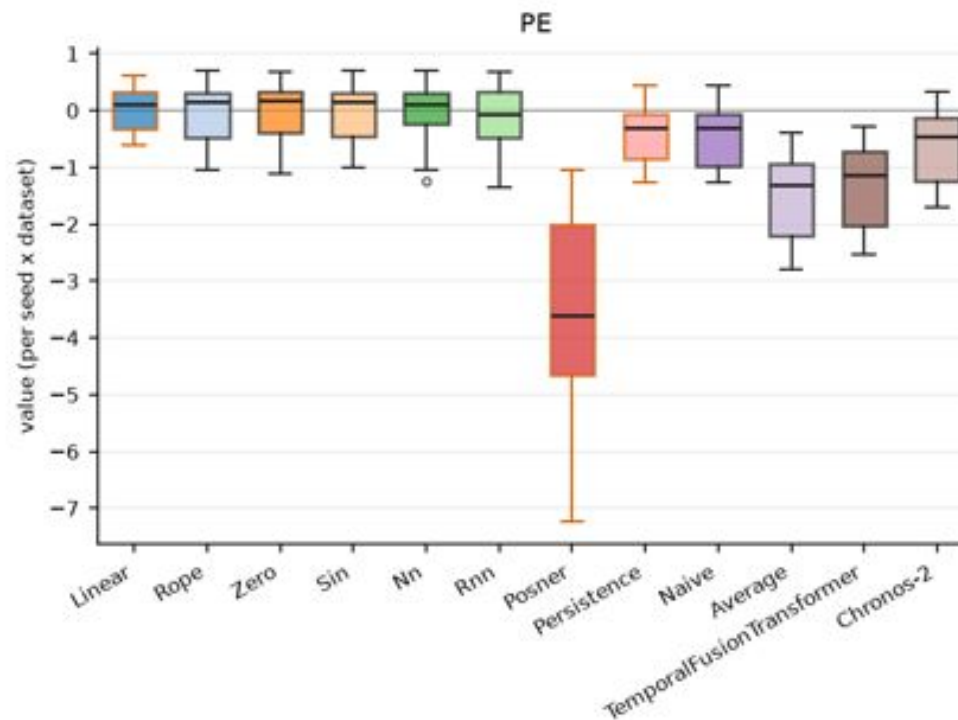


Results PE

t+6

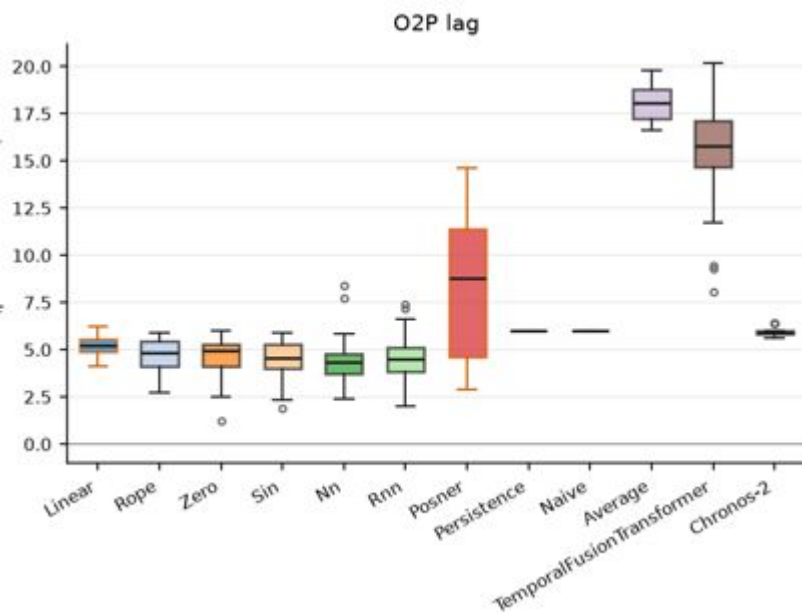


t+12

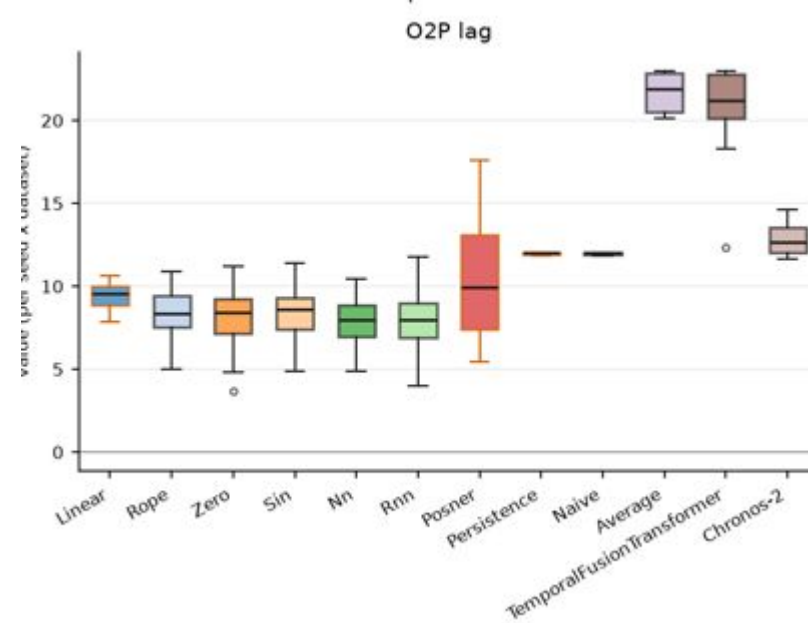


Results

T+6

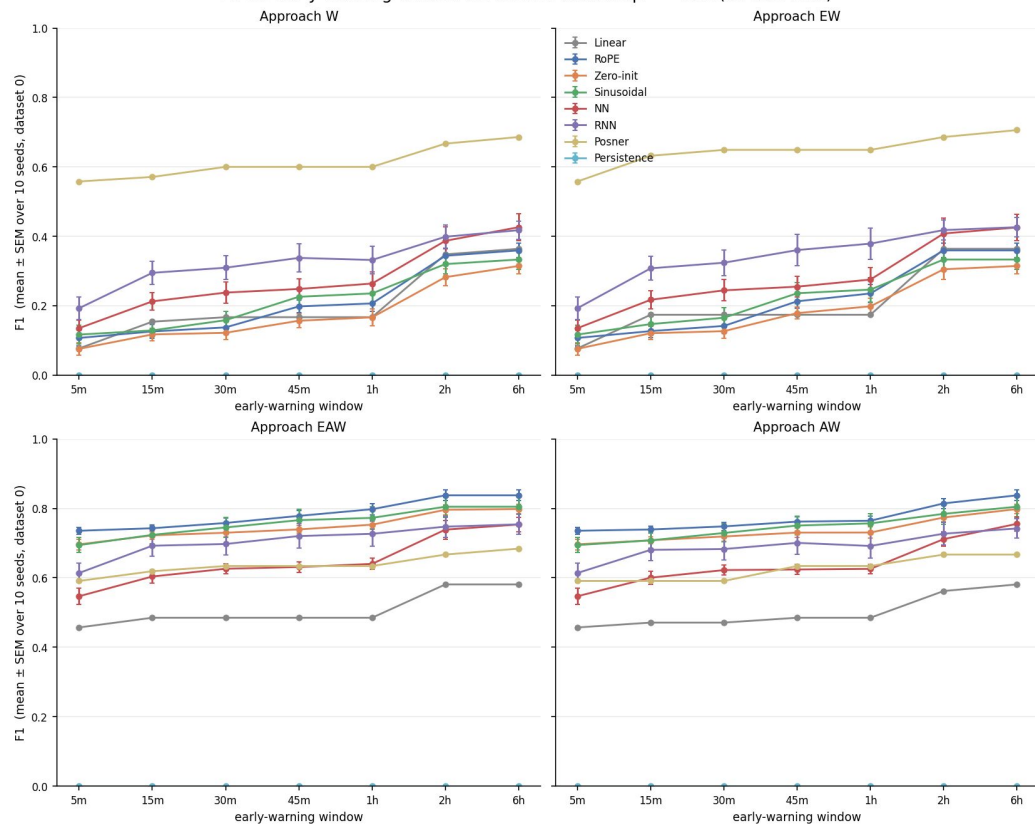


T+12



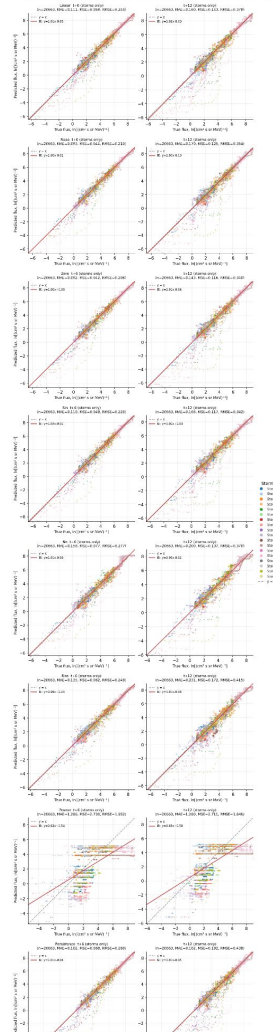
Results F1

F1 vs. early-warning window across the bootstrap — t+6 (30 min lead)

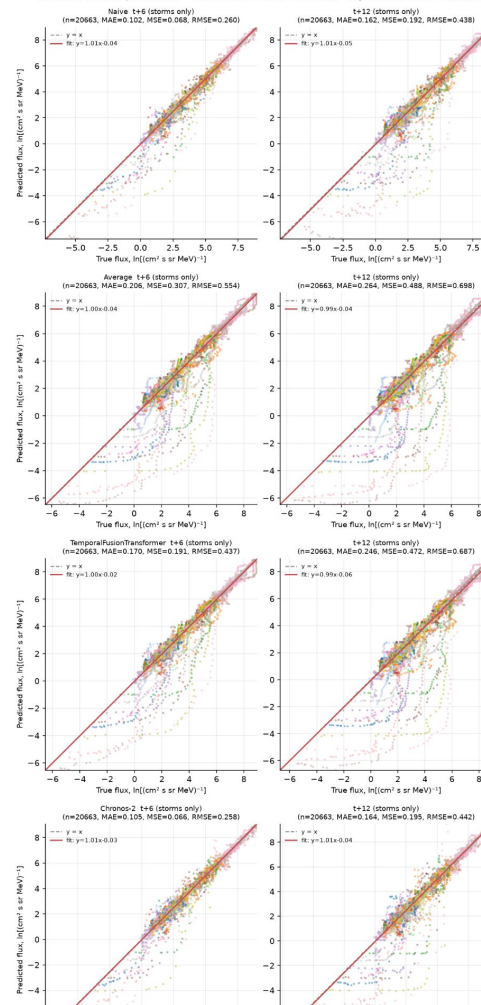


True vs Predicted

Predicted vs. true proton flux -- dataset 0, seed M1-00, storms only (t+6 and t+12 shown separately)



Predicted vs. true proton flux -- dataset 0, seed M1-00, storms only (t+6 and t+12 shown separately)



Time - Recorded for some not all in the works

Future work

Transfer forecasting model to LEO sats- GOES ACE DSCVR

Account for magnetic fields, electric fields, reconnection etc with prediction models.

proton /electron channels availability:

- **Goes 15**
 - [epead and hepad sensors](#)
- ACE
 - [EPAM](#)
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